PGCP-AI

PRACTICAL MACHINE LEARNING

Day 10: Ensemble Learning

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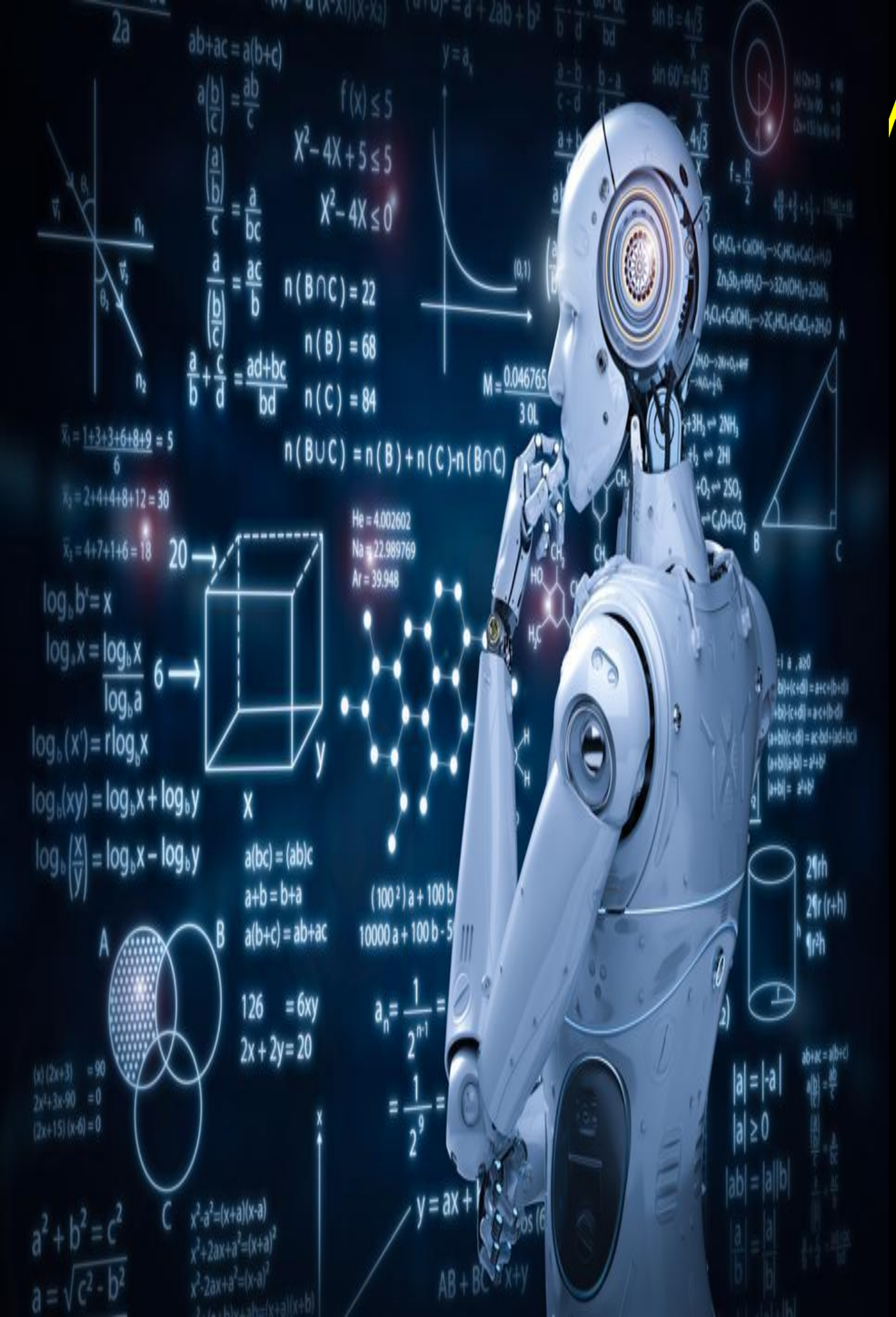


Agenda

- Ensemble Learning
- Bagging
- Boosting
- Stacking
- Random Forest

Ensemble Learning

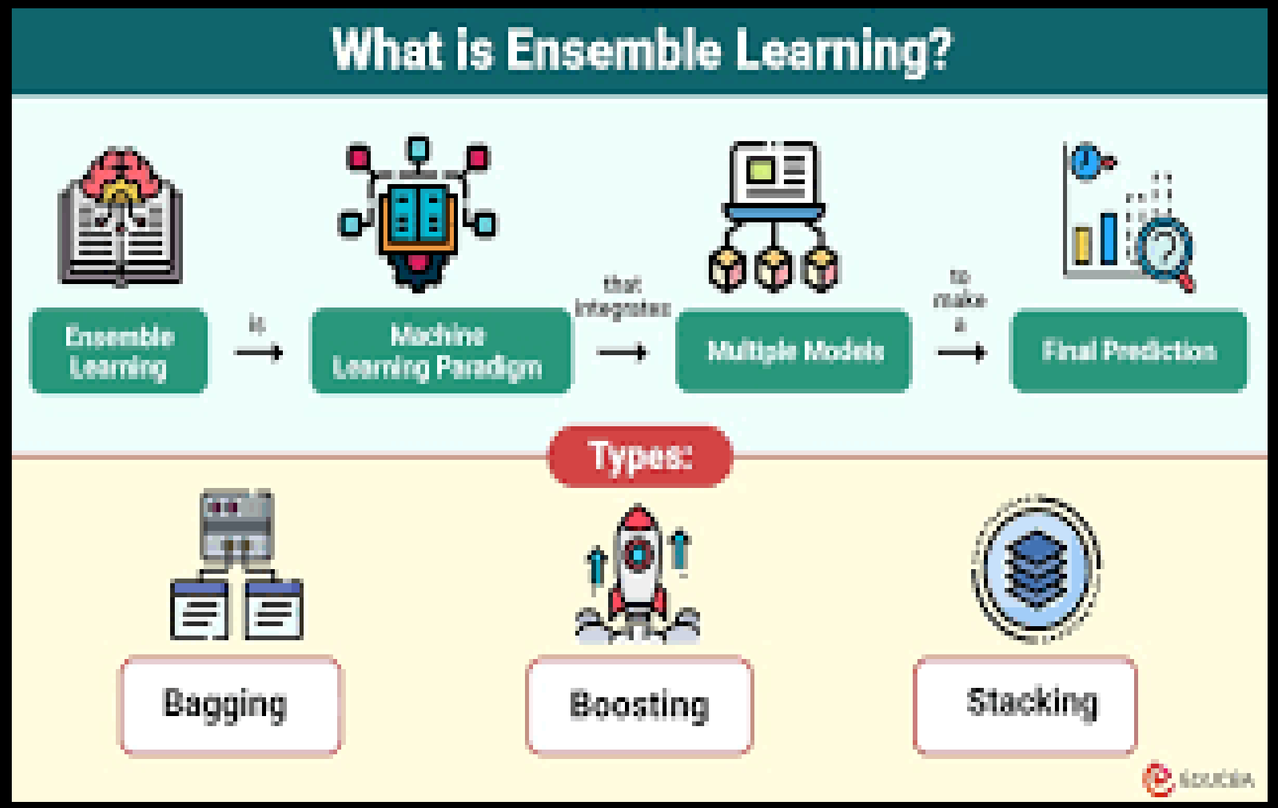
- Ensemble learning is a method in machine learning where **multiple models (often called "base learners" or "weak learners")** are combined to produce a single, stronger predictive model.

- **Key idea**
- Instead of relying on one model, you use several models and aggregate their predictions. The intuition is that different models make different errors, and combining them **reduces overall error.**

•Idea: *"Group decision is better than individual decision"*

Key Characteristics

- Multiple models trained together
- Improves accuracy and reliability
- Reduces overfitting and errors
- Works well in real-world ML problems



Why it works

- Reduces overfitting (variance)
- Can reduce bias (depending on method)
- Improves accuracy and stability
- Makes predictions more robust to noise

Why ensemble learning works

(a) Error reduction principle

Model error can be decomposed into:
Bias (error due to wrong assumptions)
Variance (sensitivity to data)
Noise (irreducible)

b) Diversity of models

Ensemble works only if models are:

- Not identical
- Make different errors

If all models are the same → no improvement.

Ensembles help by:

Reducing variance → via averaging (bagging)

Reducing bias → via sequential correction (boosting)

Bias (Systematic Error)

Definition

Bias measures how far the model's predictions are from the true values **on average.**

$$\text{Bias} = \mathbb{E}[f(x)] - y$$

It reflects **incorrect assumptions** made by the model.

Intuition

A high-bias model:

- Oversimplifies the problem
- Fails to capture patterns
- Produces similar predictions regardless of data

Example

Using a straight line to fit a curved dataset:

- The model cannot represent the true relationship
- Predictions are consistently wrong

Characteristics of high bias:

- Underfitting
- Low training accuracy
- Low testing accuracy

Models with high bias:

- Linear regression (on complex data)
- Naive Bayes
- Simple decision trees (low depth)

Variance (Sensitivity to Data)

Definition

Variance measures how much the model's predictions change when trained on different datasets.

$$\text{Variance} = \mathbb{E}[(f(x) - \mathbb{E}[f(x)])^2]$$

Intuition

A high-variance model:

- Learns noise in the training data
- Changes significantly with small data variations
- Fits training data very closely

Example

A very deep decision tree:

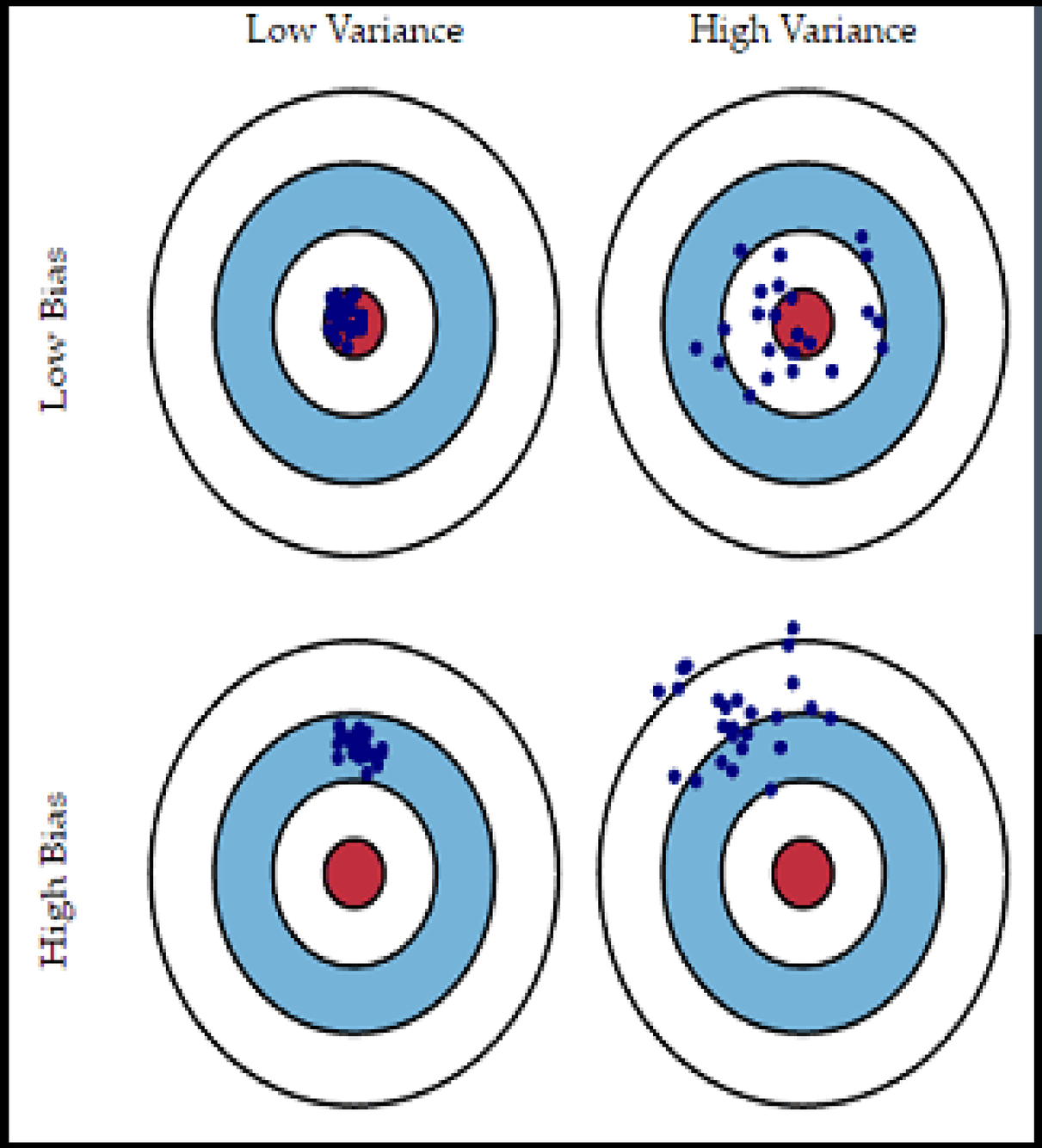
- Memorizes training data
- Performs poorly on new dat

Characteristics of high variance:

- Overfitting
- Very high training accuracy
- Low testing accuracy

Models with high variance:

- Decision trees (deep)
- K-Nearest Neighbors (low k)
- Complex neural networks



Total prediction error

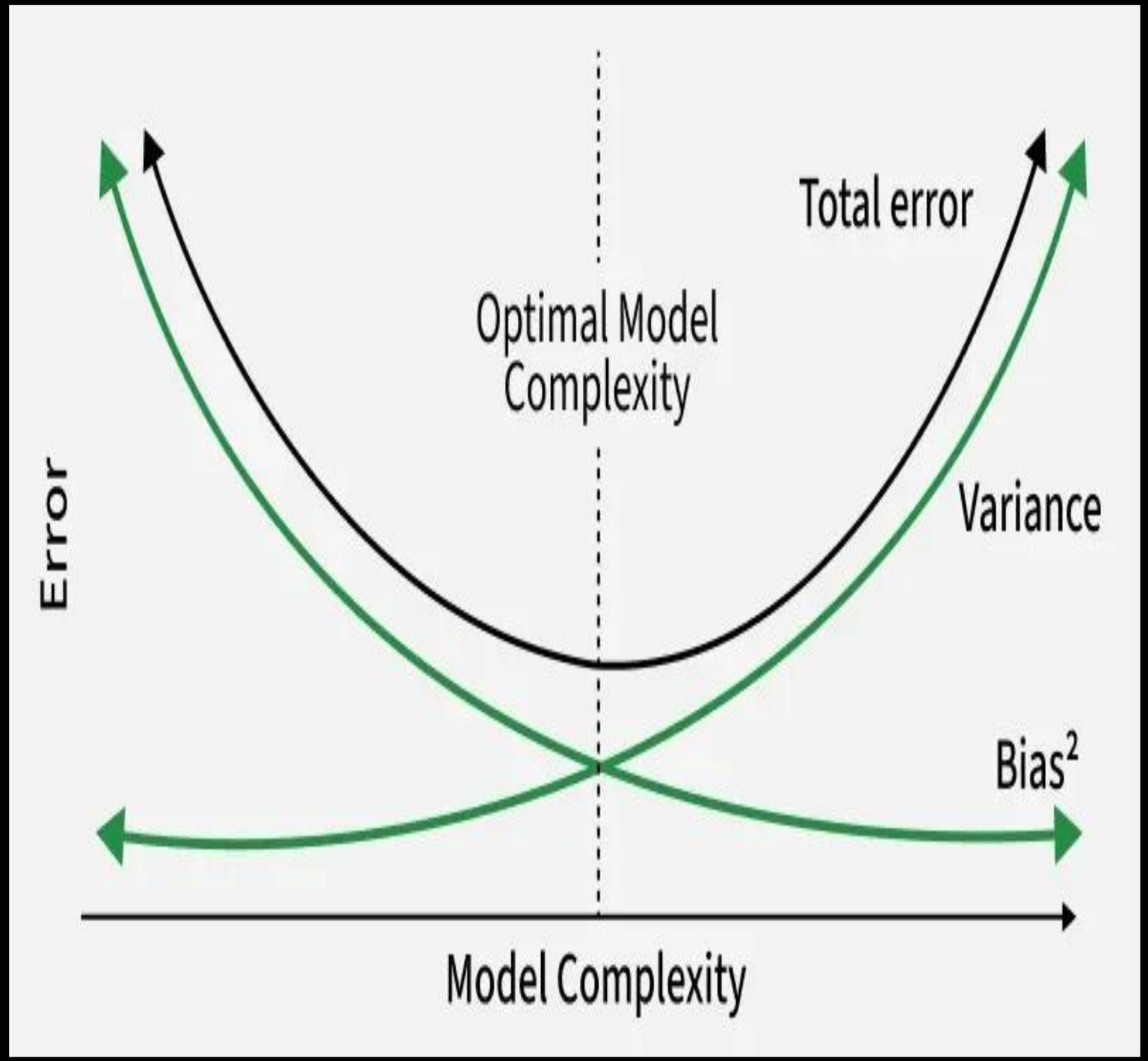
For a model, the expected error can be decomposed as:

$$\text{Error} = \text{Bias}^2 + \text{Variance} + \text{Noise}$$

- **Bias^2** → error due to wrong assumptions
- **Variance** → error due to sensitivity to data
- **Noise** → unavoidable randomness in data

As model complexity ↑

Bias ↓
Variance ↑
Optimal point = minimum total error



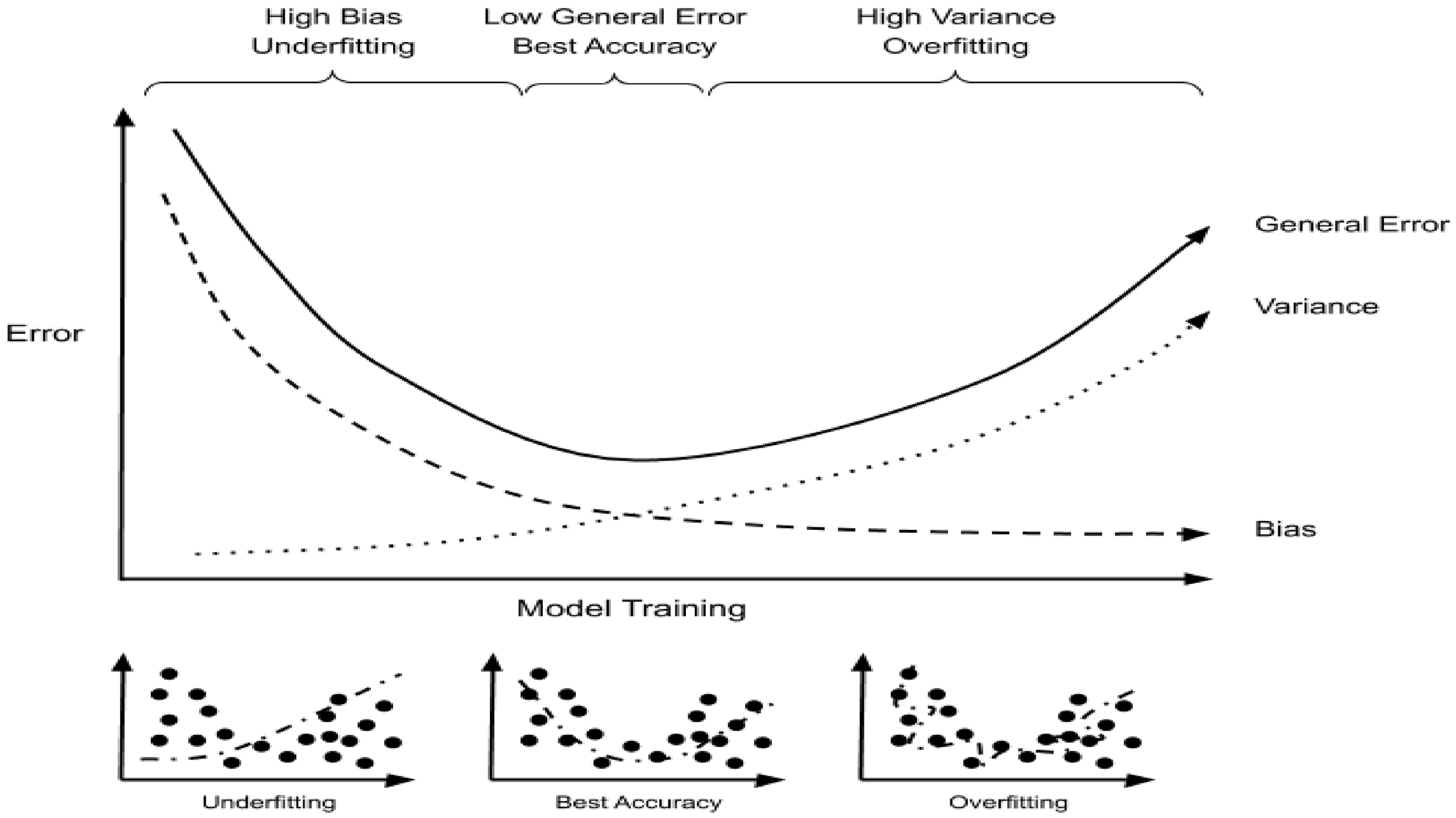
Bias vs Variance Tradeoff

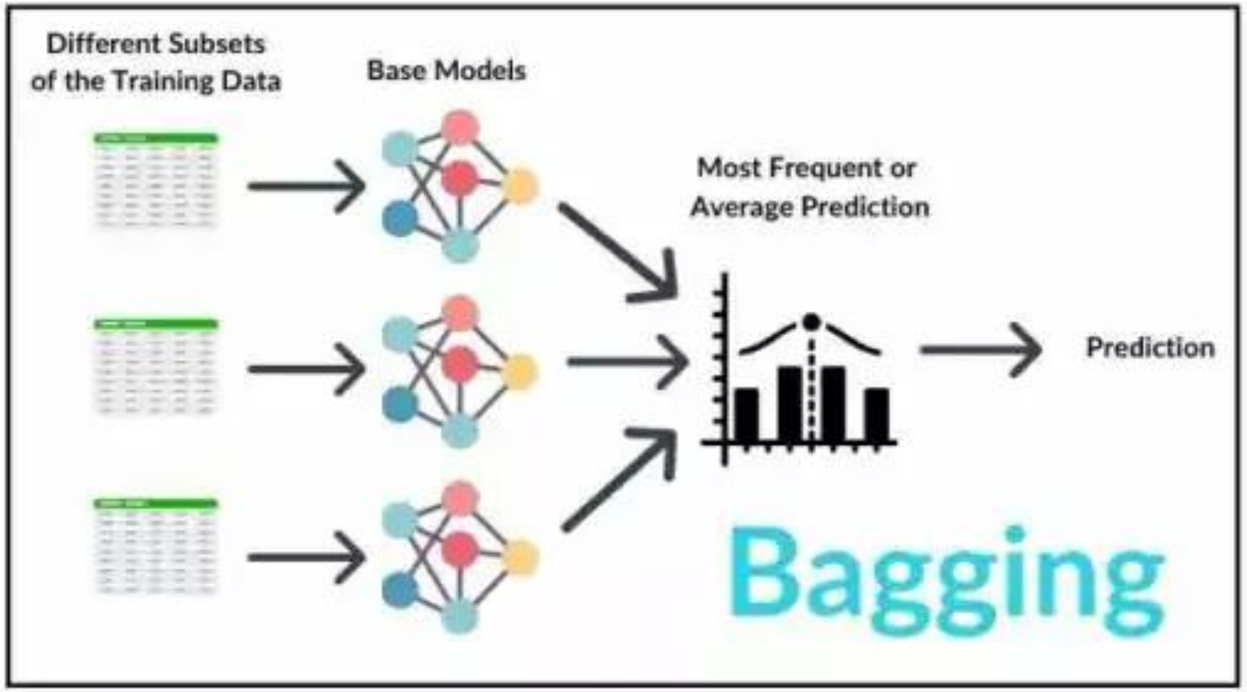
Improving one often worsens the other.

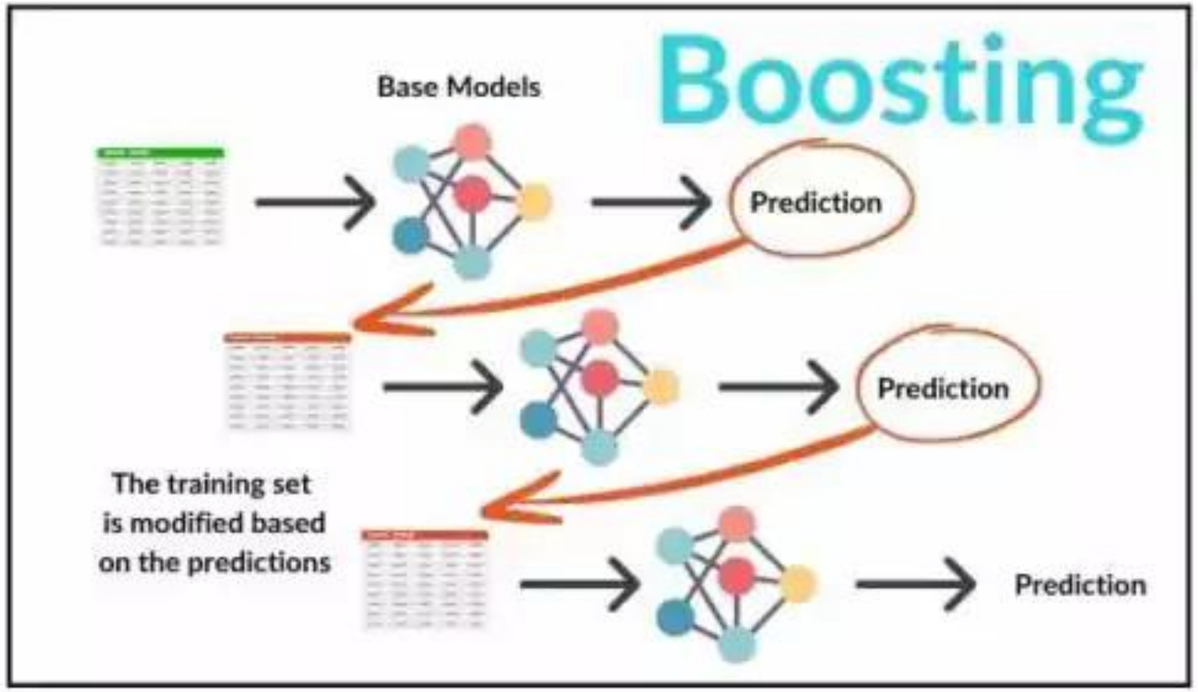
Relationship:

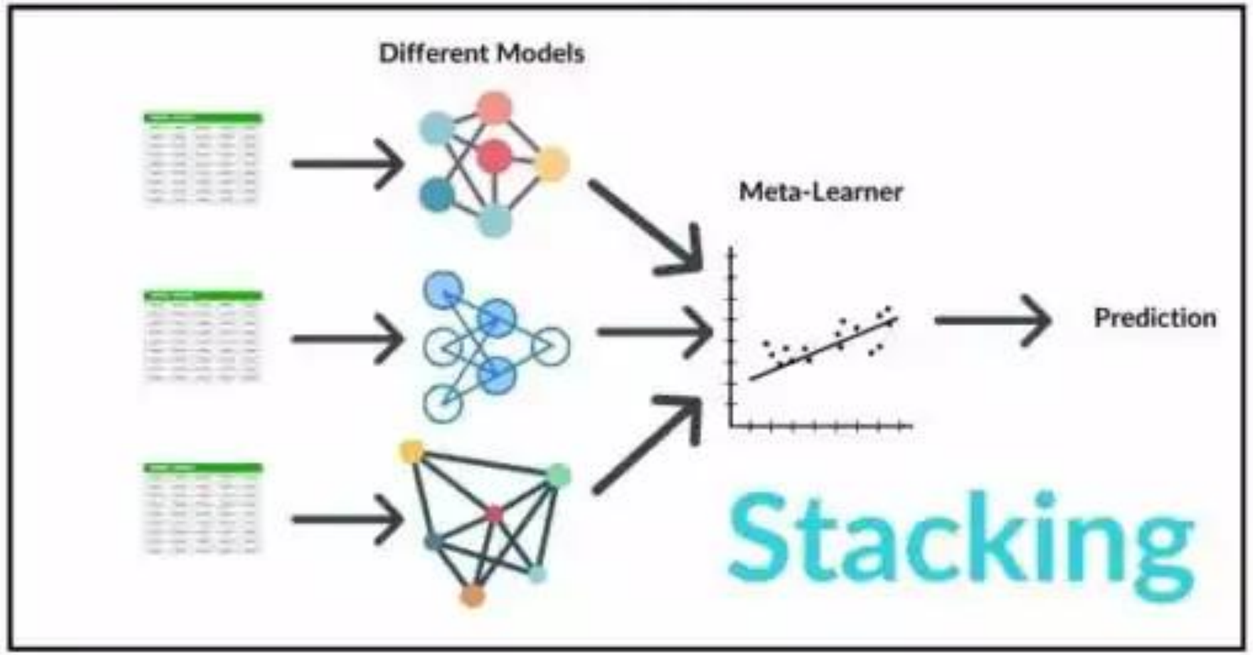
Simple models →
High bias, low variance

Complex models →
Low bias, high variance


High Bias
Underfitting
Low General Error
Best Accuracy
High Variance
Overfitting
Error
General Error
Variance
Bias
Model Training
Underfitting
Best Accuracy
Overfitting


Different Subsets
of the Training Data
Base Models
Most Frequent or
Average Prediction
Prediction
Bagging


Base Models
Boosting
Prediction
Prediction
The training set
is modified based
on the predictions
Prediction


Different Models
Meta-Learner
Prediction
Stacking

1. Bagging (Bootstrap Aggregating)

- Trains multiple models independently on different random samples of the dataset (with replacement)
- Final prediction is usually an **average** (regression) or **majority vote** (classification)

- **Example:** Random Forest
  - Many decision trees trained on different subsets
  - Reduces variance significantly

2. Boosting

- Models are trained sequentially
- Each new model focuses on correcting errors made by previous ones
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- **Examples:**
  - AdaBoost
  - Gradient Boosting
  - XGBoost

- **Key idea:** Give more importance to difficult cases over time

3. Stacking (Stacked Generalization)

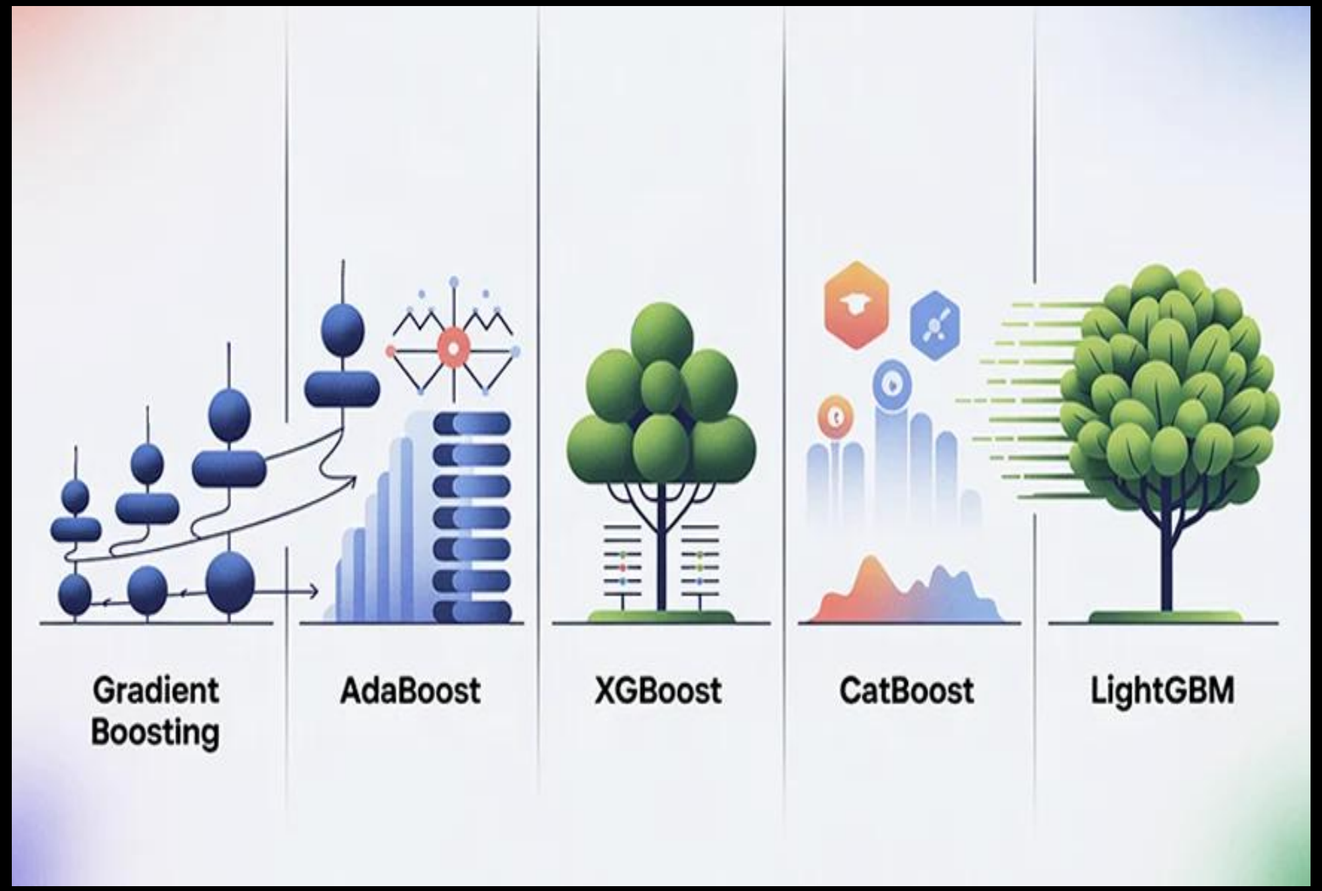
- Combines different types of models (e.g., SVM, decision tree, logistic regression)
- A “meta-model” learns how to best combine their predictions

Comparison of techniques

Method	Training Style	Goal	Strength
Bagging	Parallel	Reduce variance	Stable models
Boosting	Sequential	Reduce bias	High accuracy
Stacking	Hybrid	Combine models	Flexible

Types of boosting:

- (a) AdaBoost
  - Assigns weights to training samples
  - Misclassified points get higher weightUses weak learners (often decision stumps)
- (b) Gradient Boosting
  - Uses gradient descent to minimize loss
  - Each model fits residual errors:
  - Residual=y−F(x)
- (c) XGBoost
  - Optimized version of gradient boosting
  - Faster, regularized, highly accurate
  - Key properties:Reduces bias and variance
  - Sensitive to noise
  - Not easily parallelizable



Aspect	Bagging	Boosting	Stacking
Basic Idea	Train multiple models on random subsets of data	Train models sequentially, focusing on errors	Combine different models using a meta-model
Training Style	Parallel	Sequential	Hybrid (base models + meta-model)
Data Sampling	Bootstrap sampling (with replacement)	Uses full data but re-weights samples	Uses full dataset (sometimes split for meta-model)
Handling Errors	Each model is independent	Each model corrects previous errors	Meta-model learns from base model outputs
Overfitting	Reduced	Can reduce but may overfit if not tuned	Depends on models used
Complexity	Simple	Moderate	High
Interpretability	Medium	Low	Very low
Examples	Random Forest	AdaBoost, Gradient Boosting, XGBoost	Stacked models (e.g., RF + SVM + NN)

Wine Quality

1. Overview
   - It contains results of chemical analysis of wines derived from three different cultivars (classes).
   - The task is to classify a wine sample into one of the three categories based on its features.
2. Dataset characteristics
   - Number of samples: 178
   - Number of features: 13
   - Number of classes:

3Each class represents a different type of wine.

THANK YOU